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ANOMALOUS PALEOMAGNETIC POLES FROM LATE MESOZOIC DOLERITES FROM SPITSBERGEN *

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Late Jurassic-Early Cretaceous dolerites from Spitsbergen have normal and reversed directions of magnetization at low temperatures. After thermal demagnetization, all specimens are reversed. This reversed direction is interpreted as resulting from magnetization during a period of reversed field polarity. The dispersion in the directions of magnetization after cleaning is attributed to incomplete removal of a secondary magnetization dominant at low temperatures. This secondary component is in part related to the magnetic properties of original titanomagnetite in the dolerites, and in part to post-intrusion alteration. A weighted mean, computed from directions of magnetization which remain reversed, yields a paleomagnetic pole for the dolerites in the Atlantic region at 2.0°E, 60.1°N (δ_p = 16.1°, δ_m = 16.8°). This is considerably removed from poles for other northern hemisphere rocks of the same age.

1. INTRODUCTION

The geology of Spitsbergen has been investigated on a number of expeditions under the direction of W. B. Harland of the Sedgwick Museum, Cambridge. Collections for paleomagnetic purposes have been made during the various geological surveys [1]. As part of the 1962 Cambridge Spitsbergen Expedition, a collection was made from some late-Mesozoic dolerite sills in the Isfjord region of Vestspitsbergen [2]. Preliminary results have been reported previously [3, 4] without conclusions. This account presents a first attempt to apply demagnetization techniques to the dolerites.

Throughout a wide area of Spitsbergen, dolerite sills and bodies are emplaced in sub-horizontal, late Paleozoic and Mesozoic sediments. The sills are laterally extensive, of varying thickness, and frequently transgressive. Recent structural [5] and potassium-argon [6] studies at Cambridge have provided reasonable evidence for a late Jurassic–early Cretaceous age of intrusion (maximum K/Ar age 149 ± 17 m.y.). One hundred and five oriented blocks of dolerite were collected from a number of sites at 7 locations over an area of 1500 sq. mi. in Isfjorden (table 1).

2. MAGNETIC MEASUREMENTS

The directions of the natural remanent magnetization were measured on an astatic magnetometer at Cambridge University [7], and corrected for the gentle regional dips observed in the surrounding sediments. These corrections were never more than 6°. Of 113 cores measured initially from 105 blocks, 65 had steep, downward dipping inclination (*normal*), 32 had upward dipping inclinations (*reverse*) and the remainder were

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Table 1
Details of sampling areas.

Location	Geographic coordinates		Age of country sediments	Attitude of sediments		K/Ar age of dolerite (after Gayer et al. [6])
	Longitude	Latitude		Strike E of N	Dip	
1. Ekmanfjellet	14.3°E	78.7°N	M. Trias	154°	6°SW	
2. East coast of Bertilryggen	14.5°E	78.6°N	M. Permian	127°	4°SW	
3. Tschermakfjellet	15.3°E	78.5°N	U. Trias	92°	4°S	135 ± 15 m.y.
4. Norskedalen	15.8°E	78.5°N	M. Trias	78°	4°S	125 ± 12 m.y.
5 Anservika	16.4°E	78.4°N	M. Permian	135°	4°SW	110 ± 10 m.y.
6. Botneheia	16.4°E	78.3°N	U. Trias	87°	7°S	
7. Diabasodden	16.1°E	78.3°N	U. Trias	90°	4°S	149 ± 17 m.y.

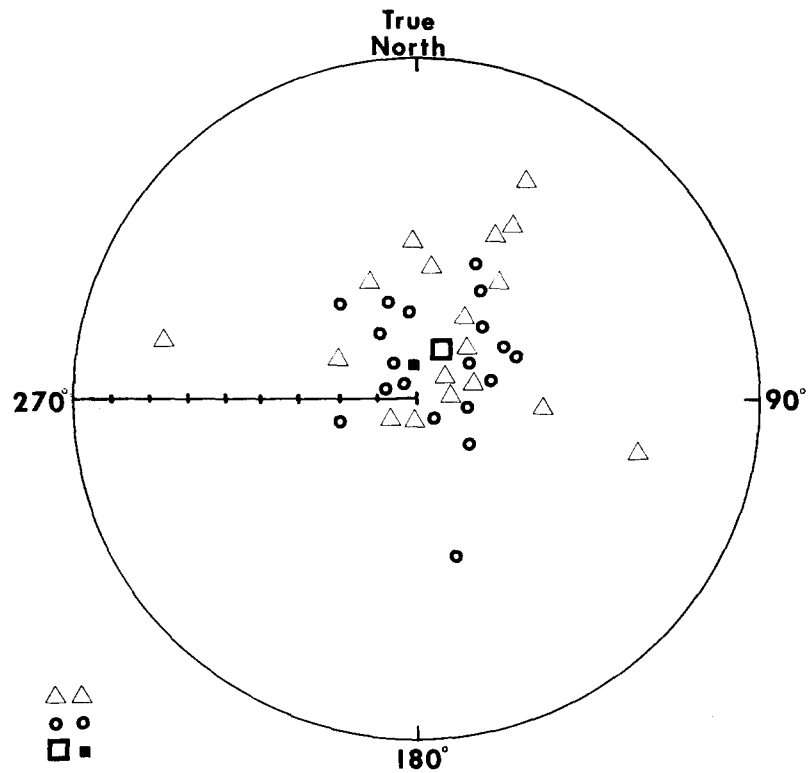


Fig. 1. Equal area projection of directions of magnetization for each specimen after thermal demagnetization. All are north poles upward seeking.

△ specimens which were initially normal
○ specimens which were initially reverse
□ mean for all specimens giving each individual weight (table 2, no. b)
■ present field.

intermediate. *Reverse* directions predominated in the fine grained dolerite along sill margins. After storage for 9 months, remeasurement of the cores showed rotation of the magnetic vector of up to 60° for some of the initially *normal* specimens, but less than 10° for the *reversed* specimens.

Thermal demagnetization was used in an attempt to retrieve the primary direction of magnetization in the dolerites. 37 specimens were selected (18 were *normal*), and demagnetized at Imperial College, London, using the continuous-measurement, furnace magnetometer of Wilson [8]. Measurements were made progressively at 70°C intervals up to that temperature at which the moment was undetectable. This depended on the sensitivity of the magnetometer (1×10^{-6} emu/g) and was not necessarily the Curie temperature.

At temperatures above 400°C the directions of magnetization of all the specimens were reverse (fig. 1). In some cases it was difficult to determine when movement of the magnetic vector had stopped, because of the small moment which remained at temperatures above 500°C. In part, the dispersion of the final directions of magnetization is due to this uncertainty.

3. DIRECTIONS OF MAGNETIZATION OBTAINED

At low temperatures, two thirds of the specimens showed *normal* directions of magnetization. The magnetization of all the specimens measured at high temperatures is *reverse*.

Even after thermal demagnetization, there is still some dispersion in the directions of magnetization (fig. 1). It is believed that this is due to the weak moment remaining at high temperatures, and is a consequence of incomplete removal of the secondary magnetization, which in the initially normal specimens, is dominant at low temperatures. This is evidenced by fig. 2, which shows the proportion of the initial total moment remaining after thermal demagnetization to 500°C. The moment removed from initially normal specimens was up to an order of magnitude greater than that from initially reversed specimens.

Fig. 2 shows that at 500°C about 30% of the initial moment remains in specimens initially reversed. Although some of the initially normal specimens show this characteristic, most possess only 5–20% of their

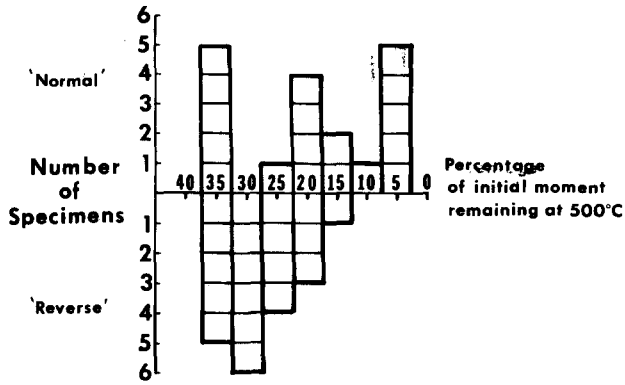


Fig. 2. Percentage of initial moment remaining at 500°C for specimens which were initially normal and those initially reverse.

initial moment at 500°C. This probably indicates that much of the moment removed to 500°C is the masking secondary magnetization.

If it is assumed that the primary thermoremanence is present in the same ratio in both normal and reversed specimens, then the secondary component must account for up to 75% of the initial moment in some of the normal specimens.

More extensive studies are required before it is known whether the secondary magnetization is of chemical or viscous origin, since it is evidently not present in all specimens to the same extent.

In part this magnetization is a post-intrusion effect. Field observations and opaque mineral petrography indicate that the dolerites have undergone moderate alteration. It has been suggested [6] that the range in K/Ar ages (110 m.y. – 149 m.y.) is due to this alteration, but it may also be related to depth of burial (table 1). Titanomaghemite is abundant along the grain margins of both homogeneous titanomagnetite, and titanomagnetite with exsolved ilmenite lamellae.

Analyses with a high field Curie balance show that the secondary magnetization is also related to the original composition of the titanomagnetite phase. Specimens initially *normal* have ill-defined Curie temperatures in the range 480°C to 560°C. Those which were initially *reverse* have more distinct Curie temperatures in the range 550°C to 600°C. Both types of Curie point curve were completely reversible.

A thermoremanence given to several specimens by cooling them in air from temperatures above 700°C

Table 2
Mean directions of magnetization after partial thermal demagnetization.

Location	N	D	I	k	α_{95}	Paleomagnetic pole	
						Longitude	Latitude N
1	4	348.3°	−79.6°	26.4	18.2°	22.0°E	58.8°
2	7	272.0°	−87.7°	5.4	28.6°	36.7°E	77.5°
3	5	19.6°	−49.5°	26.0	15.3°	2.6°W	19.4°
4	5	9.1°	−73.3°	50.8	10.8°	8.8°E	47.6°
5	5	69.1°	−76.1°	34.4	13.2°	34.6°W	57.8°
6	6	151.6°	−79.7°	12.3	19.8°	105.8°W	79.0°
7	5	21.4°	−63.6°	42.7	11.8°	2.0°W	34.2°
(a) Location means	7	24.6°	−76.4°	21.6	13.3°	2.1°W ($\delta_p = 22.8°$)	53.4° ($\delta_m = 24.6°$)
(b) Unit wt to each specimen	37	26.2°	−77.3°	10.5	7.6°	2.9°W ($\delta_p = 13.3°$)	55.1° ($\delta_m = 14.3°$)
(c) Unit wt to specimens remaining reverse on thermal demagnetization	19	68.9°	−80.2°	14.9	8.6°	2.0°E ($\delta_p = 16.1°$)	60.1° ($\delta_m = 16.8°$)

N is number of specimens; D is declination, east of N; I is inclination up; k is estimate of precision; α_{95} is semi-angle of cone of 95% confidence; δ_p and δ_m are the radial and transverse semi-angles of the oval of 95% confidence.

in a field of 0.5 Oe., always resulted in a normal direction of magnetization.

All the above data support the conclusion that these rocks were initially magnetized by a field of reverse polarity, rather than by spontaneous self reversal within the rock.

4. FISHER's STATISTICS AND PALEOMAGNETIC POLES

Initially, Fisher's statistics [9] were applied to the data in two ways: (a) to each location separately and (b) giving all the specimens individual weight. The means from either scheme differ by only a few degrees (table 2). In both cases they are divergent from the present field. Paleomagnetic poles computed on a dipolar model are also listed in table 2.

The estimates (k) of Fisher's true precision parameter κ range from 5.4 to 50.8, and the semi-angles of 95% confidence vary from 10.8° to 28.6°. Thus data might be discarded using either an F-distribution test on k values, or the arbitrary criterion that α_{95} should not be greater than 25° [8, p. 102]. However, elimination of poor data by these methods does not significantly affect the overall mean. Until more data are obtained from other stability studies, no selection of the data on a statistical basis seems warranted at the present time.

However, an arbitrary weighting procedure has been used to compute a weighted mean (table 2, no. c) using only those specimens which *remain* reversed during thermal demagnetization. Use of these data removes much of the dispersion observed at high temperatures, due to incomplete removal of the secondary magnetization. If it is accepted that the range in radiometric ages is due to later alteration, then there is no geologic evidence for several periods of intrusion and thus there is little possibility of introducing a sampling bias by using this weighting procedure. This weighted mean gives a paleomagnetic pole at 2.0°E, 60.1°N. This differs from the unweighted mean by only 7°, which is still within the confidence region of either mean.

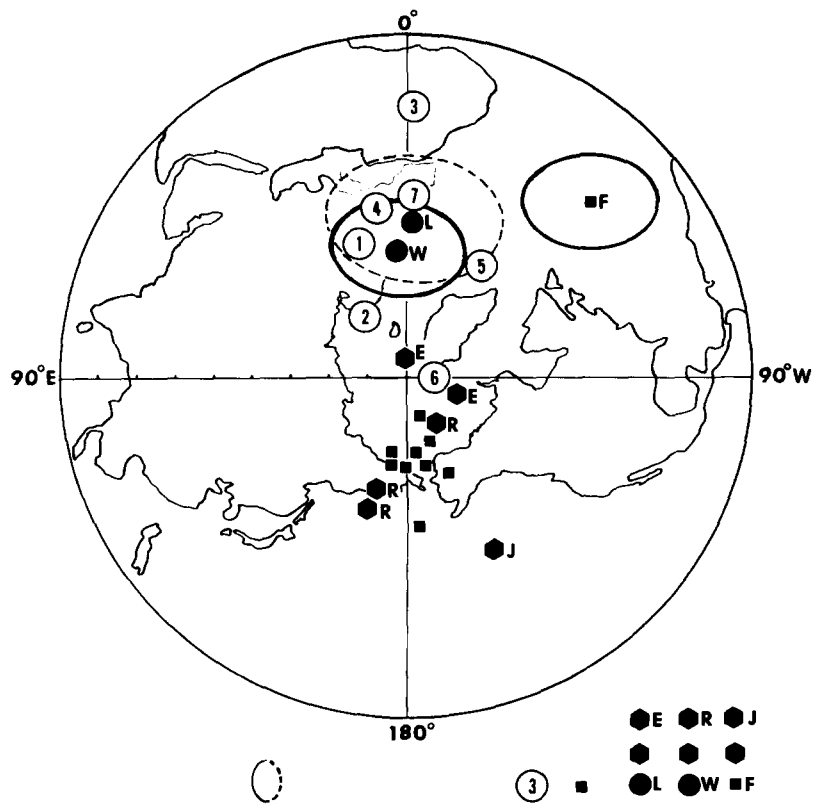


Fig. 3. Equal area projection of the northern hemisphere showing paleomagnetic poles obtained for Upper Jurassic and Cretaceous rocks from the northern hemisphere.

Data from Irving [10], Hanna [11], and Grommé, Merrill and Verhoogen [12]:

- E England
- R Russia
- J Japan
- North America
- F Franciscan Formation, California

For Spitsbergen dolerites: this paper

- ③ Locality means
- L● Overall locality mean (table 2, no. a)
- W● Weighted mean (table 2, no. c) giving individual weight to all specimens which remain reverse on thermal demagnetization.
- Ovals of confidence at 95% probability level.

5. SIGNIFICANCE OF PALEOMAGNETIC POLES

Fig. 3 compares paleomagnetic poles for location means for the Spitsbergen dolerites (numbered) and the weighted mean, with other Upper Jurassic–Lower Cretaceous poles from rocks in the northern hemisphere [10].

Poles for Cretaceous rocks from Europe and Asia are few and widely dispersed. In the main, Cretaceous poles from northern hemisphere rocks have longitudes from 90°E to 270°E. Subsequent published poles [11, 12] and data from 14 Cretaceous locations in North America [13] confirm this generalization.

The only other pole in the north Atlantic for this time, has been obtained for some spilites and diabases from the Franciscan Formation (Upper Jurassic?) in California [14]. Both the California and Spitsbergen poles can be interpreted as due to rotation of the sampling areas. A rotation of 160° would be then required for Spitsbergen. This seems excessive and there are

geological arguments against this amount of rotation [15].

The alternative explanation is that these two anomalous poles in the north Atlantic represent short period geomagnetic fluctuations. Whether these are due to localized anomalies or reflect a world wide event can only be checked by more data from other continents. In this respect it is interesting that recomputation of the Spitsbergen pole using coordinates obtained from a computer fit of the continents around the Atlantic ocean, and bearing in mind that this probably gives a pre-Jurassic model [16], shows that the anomalous Spitsbergen and California poles move 25° toward each other, becoming separated then by 20°.

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